MATCHA: Mobility Augmenting Teleoperation-Controlled Humanoid Arm

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***Abstract*—Grasping is difficult for humans with disabilities. We examine the performance of humanoid robots at replicating grasps with restricted fingers via teleoperation using a virtual reality headset. We contribute an updated hand controller and dataset collection library for the open-source AVP Teleoperate project. We use this library to collect RGB and depth recordings of restricted grasps. Finally, we design and test controllers for interpolating finger poses, using our recordings as feedback to improve restricted grasping performance. We demonstrate accuracy rates of $58-62\%$ in real-world restricted grasping performance across a set of basic tasks using our finger controllers.**

***Index Terms*—Humanoid, Virtual Reality, Teleoperation, Disability, Underactuation**

I. Introduction

With the integration of humanoids into workplaces, there is a growing need for humans to control/direct the hardware in order for the desired action to be performed. Teleoperation has been identified as a prominent solution for these situations. Most notable is the performance of repetitive movements to train machine learning models or to perform difficult tasks using teleoperation that involve dangerous environments. The main forms of humanoid teleoperation include collecting information via data gloves or virtual reality (VR) hand tracking, both of which require full range of motion of the hands to produce the intended results.

However, a significant portion of the workforce lacks full hand functionality. Individuals with conditions such as osteoarthritis, traumatic injuries, or congenital limb differences may experience restricted finger motion or possess fewer fingers. These impairments limit the usability of standard teleoperation systems, which are typically designed for the full use of five fingers and may be unusable for someone with only two.

For the purpose of focusing our study, we chose one type of hand-related disability, missing fingers, to emulate in our testing and construct a humanoid teleoperation system via VR hand tracking for these disabled individuals.

II. Previous Work

Prior work in robotic teleoperation and grasp synthesis has laid a foundation for enabling remote manipulation in human-robot collaboration contexts. Systems like Unitree's `avp_teleoperate` [1] and its predecessor OpenTeleVision [4] have demonstrated the feasibility of VR-based control for humanoid robots, allowing human operators to control full-body motion and achieve whole-body coordination in immersive environments. In the domain of grasping, studies such as Feix et al.'s GRASP taxonomy [2] have established a structured classification of human hand grasps, serving as a reference for mapping robotic hand behaviors to human-like motions.

Fig. 1. Experimental setup with Unitree H1-2 positioned parallel to table edge with external RealSense camera positioned 3 feet (center aligned) in front of the humanoid's front torso plate pointed at the robot's dextrous hand workspace. The Unitree H1-2 is a humanoid robot with 27 degrees of freedom and integrated camera with LIDAR.

In neuroscience and biomechanics, researchers have shown that human hand motion can be largely explained by a small number of joint coordination patterns, or synergies [3]. These findings have influenced control strategies that simplify high-dimensional input spaces.

Recent learning-based approaches like CoGrasp [6] have advanced the use of 6-DoF grasp generation in collaborative settings, leveraging neural models to predict feasible grasp poses in cluttered environments. These efforts collectively motivate our approach, which integrates immersive teleoperation with simplified control strategies, and aims to accommodate users with varying physical capabilities while retaining dexterous manipulation capabilities.

III. Methods

To emulate disability/hand impairment, we restrict our teleoperator to using two fingers (their thumb and index finger) in performing the grasps Fig 1. We propose two different strategies for controlling the humanoid digits that aren't directly actuated: Pose Matching and Force Matching.

A. Pose Matching

In this method, we attempt to use the unactuated digits as "support fingers". Given information about two fingers from our teleoperator, we identify the angles set for the non-thumb finger in the humanoid. We then match this value in the remaining, unknown fingers.

B. Force Matching

We reconsider the idea of using "support fingers". In this case, we match the average force felt by the sensors in the "tip" portion of the index finger to the remaining fingers. This is accomplished using the tactile feedback provided by the Inspire Hands. In this hardware, there are patches of tactile sensors throughout the inner hand positioned at common areas to experience force from an object being interacted with. The "tip" portion we take information from corresponds to the upper fingertip near the nail as seen in Fig 2.

Fig. 2. Teleoperator's finger positioning during testing

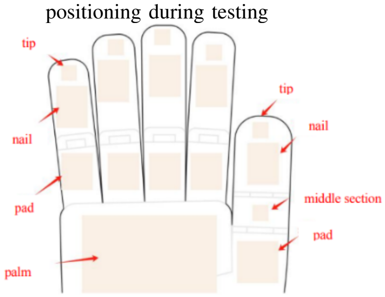

Fig. 3. Pressure Sensor Map in Inspire Hands

IV. Experimental Setup

A. Hardware and Setup

The following key hardware was used in our experimentation:

- **Humanoid Robot:** Unitree H1-2 (6-DOF arms)
- **Robotic Hand:** Inspire RH56DFTP (5 fingers, 6-DOF acutation, tactile sensors)
- **Camera:** Intel RealSense D435 (depth + RGB)
- **Compute Unit:** Razer 15 Laptop
- **Grasped Objects:** The robot was tested on a diverse set of everyday items, including:
  - Pear
  - Grapes
  - Water bottle
  - Tissues
  - Stapler
  - Box
  - Pliers
  - Mug
  - Pen
  - Measuring Tape

The items chosen were selected to demonstrate grasping on varying types (e.g. rigid, malleable, small) of regularly encountered objects.

We implemented the open source repository "avp_teleoperate" from Unitree Robotics [1] for the base system and add some additional elements to be compatible with our version of the Inspire Hands (RH56DFTP).

B. Procedure

The Unitree H1-2 humanoid robot was placed in `debug` mode and carefully positioned in a standing pose over a 40-inch tall table containing various test objects. For this experiment, the robot's locomotion was fixed, and only the upper body and arms were used for grasping. The robot was suspended on a gantry with the bottom of the feet 3 inches above the floor.

To capture the interaction environment, an Intel RealSense D435i RGB-D camera was mounted on a tripod three feet away from the robot's torso. The camera was center-aligned and oriented to provide a frontal view of the workspace, ensuring full coverage of the object interaction zone.

Each trial involved two researchers operating the system from a safe distance. One operator used a virtual reality (VR) headset and hand tracking interface to control the robot in real time, sending desired joint commands for the thumb and index finger. The second operator monitored the robot's execution and ensured all control scripts and data logging systems were functioning properly on the host computer.

Before each trial, a selected object was randomly placed in both position and orientation within the robot's reachable workspace. The experimenter then initiated the grasping attempt using one of the control strategies (pose matching or force matching). The grasp was deemed successful if the robot lifted the object at least three inches off the table without dropping it.

A sample trial setup can be seen in the video linked below:
Bottle Grasping Demo

Ten different, previously selected objects are used for this experiment with five grasp attempts being conducted for each object. This is carried out for both methods (pose matching and force matching) resulting in a total of 100 trials. We formally define a grasp attempt by the action of being in contact with

Fig. 4. Object (pear) placed randomly within the bounds of the humanoid's camera scope.

the object followed by lifting the humanoid's hand at least three inches off the table or, if grasped, by when the object is no longer touching the table.

We ensure in each trial the object to be grasped is placed in a random location and orientation within the bounds of the humanoid's camera as seen in Fig 4. This is done for grasping variability purposes, as certain placements of an object can influence the grasping approach used by the teleoperator and change the difficulty of completing a grasp.

C. Controller Logic

We implement two primary controllers for adjusting the motion of unactuated fingers based on input from the index finger: a pose-based matching controller and a pressure-based feedback controller. The pseudocode for each is provided below.

Pose Matching Controller:

This method simply copies the joint angle of a designated source finger (typically the index finger) to all other non-thumb fingers. This allows the unactuated fingers to mimic the pose of the source finger directly, assuming identical joint configurations.

Algorithm 1 Copy Source (Index) Finger Angle to All Fingers

Require: Joint values $\mathbf{q} \in \mathbb{R}^6$
Require: Source finger index s
$\textit{fingers} \leftarrow \{0, 1, 2, 3\}$ ▷ Indices for pinky, ring, middle, index
for all $i \in \textit{fingers}$ **do**
  if $i \neq s$ **then**
    $\mathbf{q}[i] \leftarrow \mathbf{q}[s]$
return q

Pressure Matching Controller:

This controller adjusts each follower finger based on the pressure sensed at its fingertip relative to the pressure at the index fingertip. If the pressure is too low, the finger curls further inward. If the pressure is too high, it straightens out. A small threshold ε is used to prevent unnecessary jitter from sensor noise.

Algorithm 2 Match Source Finger Pressure for All fingers

Require: Joint values $\mathbf{q} \in \mathbb{R}^n$, source finger index s
Require: Pressure map *pressures*, patch map *patches*
Require: Step size δ, tolerance ε
$p_s \leftarrow \textit{pressures}[\textit{patches}[s]]$
for all $i \in \textit{fingers},\ i \neq s$ **do**
  $p_i \leftarrow \textit{pressures}[\textit{patches}[i]]$
  $\Delta p \leftarrow p_s - p_i$
  if $p_s < \varepsilon$ **then**
    $\mathbf{q}[i] \leftarrow 1.0$ ▷ Fully extend
  else if $\Delta p > \varepsilon$ **then**
    $\mathbf{q}[i] \leftarrow \max(0.0,\ \mathbf{q}[i] - \delta)$ ▷ Curl more
  else if $\Delta p < -\varepsilon$ **then**
    $\mathbf{q}[i] \leftarrow \min(1.0,\ \mathbf{q}[i] + \delta)$ ▷ Straighten more
return q

V. Results

We were able to write a new joint control class for the new Inspire RH56 DFTP hands. These hands were not previously supported in AVP Teleoperate and we contributed this functionality to be able to use the tactile sensors in the hands. We were also able to bringup the AVP Teleoperate software stack from scratch on the H1-2 humanoid after extensive debugging of the local network. We set up a brand new router and set the IP addresses of all devices on the network to communicate with each other and send joint commands from the Apple Vision Pro to the humanoid onboard computer with low latency.

TABLE I
Pose Grasping Task Results with Success Counts

Object	Success #	Attempt #	Success Rate
Pear	5	5	**100%**
Grapes	3	5	60%
Pen	0	5	0%
Box	4	5	80%
Water bottle	5	5	**100%**
Tissue Pack	3	5	60%
Stapler	2	5	40%
Mug	4	5	80%
Pliers	0	5	0%
Measuring tape	3	5	60%
Total	**29**	**50**	58%

We were able to collect all of our 100 trials, 50 each with the index pose matching and index pressure matching augmented finger controllers. We aggregate our results for each in Table I and Table II. The index pose matching controller achieved an overall 58% accuracy and the pressure matching achieved 62% accuracy on success in grasping trials.

TABLE II
PRESSURE GRASPING TASK RESULTS WITH SUCCESS COUNTS

Object	Success #	Attempt #	Success Rate
Pear	5	5	**100%**
Grapes	2	5	40%
Pen	0	5	0%
Box	4	5	80%
Water bottle	5	5	**100%**
Tissue Pack	4	5	80%
Stapler	4	5	80%
Mug	3	5	60%
Pliers	0	5	0%
Measuring tape	4	5	80%
Total	**31**	**50**	62%

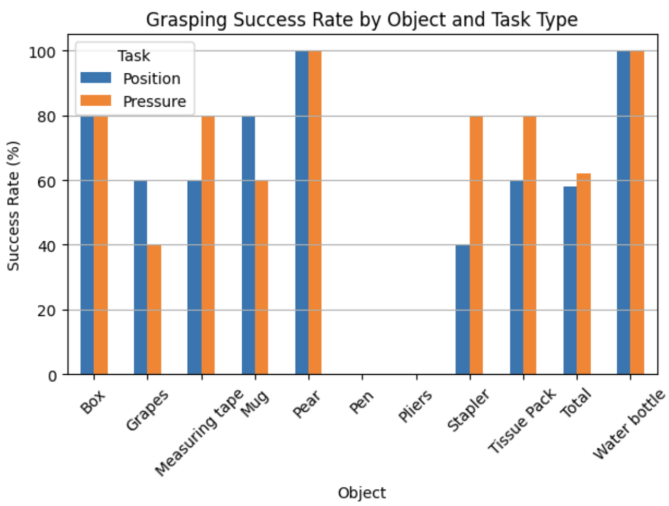

Fig. 5. Bar chart showing the relative performance of the pose matching and pressure matching controllers for each object grasping trial.

VI. DISCUSSION

Based on the success rates alone, we recognize a relatively equal capability among the methods. However, this is not the only way to measure performance and is not sufficient for understanding the method at play. We must also consider an "ease of use" factor, as our teleoperators found certain difficulties in each method that is not wholly expressed through the success rate.

The pose method had a unique struggle with the stapler due to the springs in the system. Upon grasping from the top and bottom of the object the supporting fingers would apply additional pressure to the stapler, due to the curved nature of the object's upper portion, and end up stapling. This resulted in the stapler suddenly jolting open and escaping from the grasp of the humanoid.

It is relevant to note, the force matching control did not always operate as intended. In practice, we witnessed a lot of jittery movement in the support fingers. This is due to how the code is structured - we set the support fingers to increase its angle every time the force experienced in the respective fingertip is less than the one in the index fingertip. Similarly, if the force is greater in the support fingers their angle decreases. We incorporate an area for error before movement using a threshold value. We believe that a reason for the undesirable movement is due to not tuning this threshold value tightly enough.

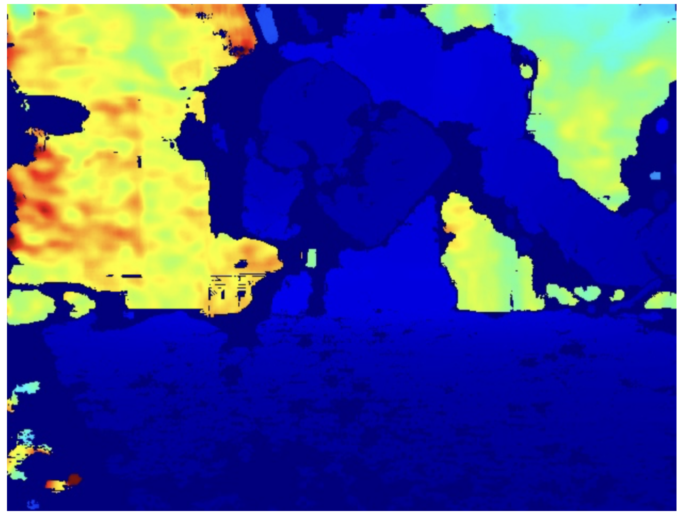

Fig. 6. Depth footage from external RealSense camera of a trial picking up a mug using the pose matching controller.

Fig. 7. RGB color footage from external RealSense camera of a trial picking up a mug using the pose matching controller.

Additionally, the tactile feedback from the Inspire Hands proved to be difficult to work with. Routinely, the force measured by picking up an object would not exceed random noise of the system. Through attempting to neglect the noise, the support fingers would not begin helping until a large amount of force was being exerted. This was not ideal as it resulted in us using more energy than necessary in addition to potentially being a safety hazard. For example, following our pear picking trials, we witnessed an incredible amount of dents in the pear's skin. Thus, this method would be less than ideal for fragile objects or life forms with pain receptors.

Furthermore, we notice that occasionally the force based method can be a detriment to our grasp attempt. This is most notable in the cases of objects that are very short. When attempting to pinch the object with the humanoid's fingers, the support fingers would bend and make contact with the table - preventing the teleoperator from moving the index and thumb closer to the table. This is directly due to the incremental movement of the controller, making small angle changes every control loop. When force is exerted on the index finger during the grasping the supporting fingers will curl in and end up pushing against the table, thus compromising our initial grip.

The additional benefit of the support fingers as opposed to a system with just the base precision grip (index and thumb) is unclear. In both methods we witnessed the remaining fingers providing support and further stabilizing the object but not contributing to the grasping strategy or key points in the grasp. Both methods struggled equally with small items like the pen and pliers since the base precision grip struggled to pinch the area near the table, thus making the supporting fingers' movement irrelevant.

VII. Future Work

While our current system focuses on basic grasping and object pickup, future efforts will aim to extend its capabilities toward more sophisticated manipulation tasks. This includes enabling tool use, in-hand object reorientation, and dual-hand grasping strategies. These enhancements would allow for greater dexterity and utility in daily assistance and teleoperation contexts.

There are several implementation improvements we'd like to make in the future to make our dataset and controllers more useful. The first improvement would be adding more external cameras to view the grasping motion of the hands from different angles to help future grasping models train better on visual features indicating a secure grip, a slipping object, or other states. We'd also like to rescale the depth range on the RealSense to increase the fidelity of depth recorded in the hand workspace. Currently as seen in the above image, the distant wall behind the robot greatly stretches the depth range and lowers the depth resolution of the actual hands and gripped object. We'd also like to collect an order of magnitude more trials for each object, using multiple human operators to control for variation in human teleop performance and multiple disabled finger configurations to examine how each finger individually contributes to grasp performance. Finally, we didn't have a chance to perform a wide sweep of parameters for our two implemented controllers that could improve the fine motion of the fingers. Examples of these parameters could be increasing the angle step size taken when compensating for reduced pressure to make the fingers adjust to new grips faster, or reducing the epsilon error in pressure measurements to make the force exerted by the fingers more even.

Besides implementation, there are some new fields we'd like to explore. One major direction is intent inference. Currently, our system assumes explicit control over all degrees of freedom, which is often unrealistic for users with reduced mobility or limited interfaces. Humans often provide only partial input when interacting with the world, yet our intent is usually still discernible. We aim to explore how to infer manipulation goals from reduced control signals. This involves recognizing object affordances, matching them to canonical hand motions, and analyzing the preceding trajectory history to understand the user's intent. Incorporating multimodal inputs—such as gaze, force feedback, or contextual cues—may also assist in disambiguating intended actions.

Additionally, we seek to leverage the concept of motor synergies to simplify control. Studies have shown that a small number of coordinated joint movement patterns (synergies) can explain the majority of natural hand motion, with roughly nine synergies accounting for over 90% of variance in hand configurations. Embedding such structure into the control policy may reduce user burden and enhance generalization.

Finally, generalization across a wide range of motor limitations remains a key challenge. Individuals may have missing digits, fused joints, or limited range of motion, requiring our system to adapt to varied and constrained input signals. A robust control framework must accommodate these differences while still enabling expressive manipulation.

Exploration of these areas will bring us closer to a system that is flexible, intuitive, and useful in real-world assistive and telepresence scenarios.

A. *Generalizing and Manipulation*

Immediately, we have to recognize the limitations of our study. The only action we tested using the proposed methods was picking up an object, albeit a variety of them. We did not explore the capabilities of these methods in manipulation, which encompasses the majority of how humans interact with objects. This restricts the use cases of the proposed methods, as such, we desire to explore additional strategies outside matching based methods.

Furthermore, in our study we pick an arbitrary hand limitation (the use of two fingers fully as opposed to five) and structure methods based on "filling in" the actions of the remaining three fingers. However, we recognize that various hand impairments exist - struggles with range of motion, actuation that doesn't include the index finger and thumb, etc. Our current implementation is catered directly to a specific disability and is not compatible with other hand impairments. In future work, we wish to identify teleoperational constraints in individuals with limited range of motion and determine a method for assistance. We seek to confirm that the proposed solution will work with the minimum set of a thumb and any finger and also to conceive new methods for a single finger or two non-thumb fingers.

B. Synergies

Prior research in neuroscience fields recognized that during both simple and skilled tasks there was a set of repeated joint movements that accounted for the majority of motion (specifically in manipulation) [3]. In future experimentation we would like to analyze if synergies (joint patterns) are found in the act of grasping specifically and if we can then use these patterns and interpolations between them to grasp any type of object. Additional research using online LLM platforms indicates that there is lots of work in this field [5].

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APPENDIX

GitHub Link: https://github.com/tsai-henry/MATCHA
Grasping Dataset: Google Drive